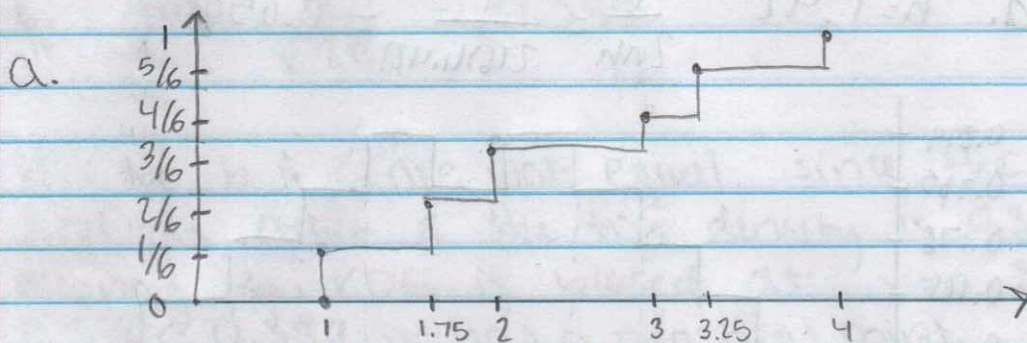
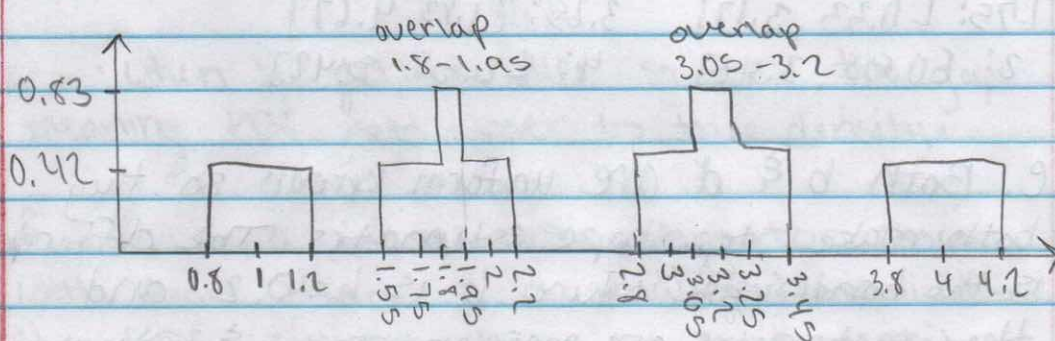


Problem 1: Data 1, 1.75, 2, 3, 3.25, 4 $n=6$



b. $h = 0.2$ $\frac{1}{2nh} = \frac{1}{2(6)(0.2)} = \frac{1}{2.4} = 0.4167$



c. $h = 1.84 \times SD \times n^{-0.2}$

$\bar{x} = 2.5$

$1 - 2.5 = -1.5 \Rightarrow -1.5^2 = 2.25$

$1.75 - 2.5 = -0.75 \Rightarrow -0.75^2 = 0.5625$

$2 - 2.5 = -0.5 \Rightarrow -0.5^2 = 0.25$

$3 - 2.5 = 0.5 \Rightarrow 0.5^2 = 0.25$

$3.25 - 2.5 = 0.75 \Rightarrow 0.75^2 = 0.5625$

$4 - 2.5 = 1.5 \Rightarrow 1.5^2 = 2.25$

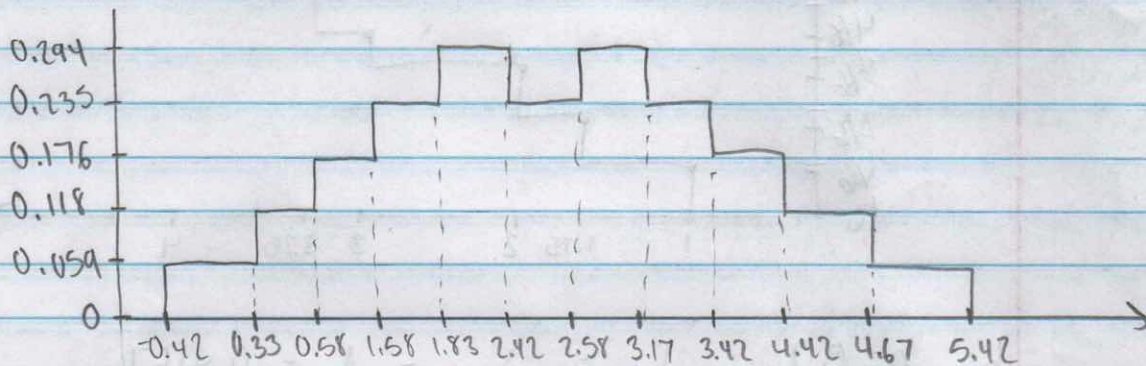
$\frac{6.125}{5} = 1.225$

$SD = \sqrt{1.225} = 1.107$

$h = (1.84)(1.107)(6^{-0.2})$

$h = 1.42$

d. $h = 1.42$ $\frac{1}{2nh} = \frac{1}{2(6)(1.42)} = 0.059$



1: $[-0.42, 2.42]$ 3: $[1.58, 4.42]$
 1.75: $[0.33, 3.17]$ 3.25: $[1.83, 4.67]$
 2: $[0.58, 3.42]$ 4: $[2.58, 5.42]$

e. Both b & d are uniform kernels so they both make step shape estimates. The difference is the bandwidth where b is $h = 0.2$ and the rectangles are more narrow & overlap less and d is $h = 1.42$ so its rectangles are wider & overlap more making the KDE more spread out.

Problem 2: Data $X = [-1, -0.5, 0, 0.5, 1]$

$X=0$
 $h=0.6$ a. $\frac{1}{2nh} = \frac{1}{2(5)(0.6)} = \frac{1}{6} \Rightarrow 3\left(\frac{1}{6}\right) = 0.5$
 $\hat{f}_h = 0.5$

b. $-0.6 < X < 0.6 = 0.451$ $2h = 1.2$
 $E\hat{f}_h(0) = \frac{0.451}{1.2} = 0.376$

c. $f(0) = \frac{1}{\sqrt{2\pi}} = 0.399$

d. No, b & c are not equal. since the KDE is 0.376 & the true density is 0.399 meaning the KDE is biased at $x=0$. Since $0.376 < 0.399$ it has a neg. bias point.

e. $-0.1 < x < 0.1 = 0.0797$ $2h=0.2$

$E\hat{f}_h(0) = \frac{0.0797}{0.2} = 0.3985$

• When h gets smaller so does the gap meaning KDE gets closer to true density.

f. $\hat{f}_h$ is a consistent estimator since with more data & a smaller bandwidth the KDE can get closer to true density.

g. The main thing we want w/ h is for it to be a balance between a small bandwidth that gives more details but is not as smooth & a large bandwidth that is smooth but can miss details.

Problem 3: $X = [5, 7, 7, 8, 10]$ $h=1.5$

$$\frac{1}{nh}$$

$$\frac{1}{5(1.5)} = \frac{1}{7.5}$$

$$0.1333$$

a. $x=6$ $\hat{f}(6) = 3(0.1333) = 0.400$

$x=6.5$ $\hat{f}(6.5) = 4(0.1333) = 0.533$

$x=7$ $\hat{f}(7) = 3(0.1333) = 0.400$

$x=8$ $\hat{f}(8) = 3(0.1333) = 0.400$

$x=8.5$ $\hat{f}(8.5) = 4(0.1333) = 0.533$

$$b. \hat{f}_h(x) = \frac{1}{nh} \sum k\left(\frac{x-x_i}{h}\right)$$

$$n=5, h=1.5, nh=5(1.5)=7.5$$

$$x=6: \frac{6-5}{1.5} = \frac{1}{1.5} = 0.667 \Rightarrow u=0.667$$

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2} \Rightarrow K(0.667) = 0.319$$

$$x=7: \frac{6-7}{1.5} = \frac{-1}{1.5} = -0.667 \Rightarrow u=-0.667$$

$$K(-0.667) = 0.319$$

$$x=8: \frac{6-8}{1.5} = \frac{-2}{1.5} = -1.333 \Rightarrow u=-1.333$$

$$K(-1.333) = 0.164$$

$$x=10: \frac{6-10}{1.5} = \frac{-4}{1.5} = -2.667 \Rightarrow u=-2.667$$

$$K(-2.667) = 0.011$$

$$0.319 + 0.319 + 0.319 + 0.164 + 0.011 = 1.132$$

$$1.132 / 7.5 = 0.151 \quad \hat{f}_h(6) = \boxed{0.151}$$

$$x=6.5: u = [1, -0.333, -0.333, -1, -2.333]$$

$$K(u) = [0.242, 0.377, 0.377, 0.242, 0.026]$$

$$\hat{f}_h(6.5) = 1.264 / 7.5 = \boxed{0.169}$$

$$x=7: u = [1.333, 0, 0, -0.667, -2]$$

$$K(u) = [0.164, 0.399, 0.399, 0.319, 0.054]$$

$$\hat{f}_h(7) = 1.335 / 7.5 = \boxed{0.178}$$

$$x=8: u = [2, 0.667, 0.667, 0, -1.333]$$

$$K(u) = [0.054, 0.319, 0.319, 0.399, 0.164]$$

$$\hat{f}_h(8) = 1.256 / 7.5 = \boxed{0.167}$$

$x=8.5$:

$$u = [2.333, 1, 1, 0.333, -1]$$

$$k(u) = [0.026, 0.242, 0.242, 0.377, 0.242]$$

$$\hat{f}_h(8.5) = 1.130/7.5 = \boxed{0.151}$$

c. The largest jump is at 0.067. The uniform kernel has a box shape so the data points either fully counts or not at all which explains the jump. The Gaussian kernel has a bell shape so its points change gradually as the distance changes which results in the KDE changing smoothly.

d. The Gaussian kernel is preferred since it gives smoother estimates (despite being an actual annoying pain to compute). The uniform is simpler to do but has more abrupt changes.